PASS-PRO

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Introduction

With our everyday lives growing more and more attached to the things we do online the need for safe, secure passwords to ensure the utmost security for our accounts, investments, and more has never been more important. There are many different types of password managers that exist within the NetSpace, however, many create passwords that are hard to remember on the spot. As such I've decided to create the PASS-PRO application to remedy that issue. No longer will any individual need to remember a random jumble of numbers, letters, and symbols- instead they'll be able to set a memorable phrase, date, or event to generate a password, one that will both be easier to recall than what's provided by competitors and be stored in a document that a user can recall at any time. The target for this app is wide spanning and encompasses everyone from the youngest of students to the oldest of engineers- everyone has some degree of an online presence after all.

The construction of the app will take a very straightforward approach. First the general outline of the program will be established with a basic user interface, after that the "guts" of the program will be built from scratch- functions defined, randomizers implemented, ect, and finally development will focus on storing the newly created passwords in a document a user can recall. Depending on how efficiently I'm able to work I think that I could add a host of newer features to the program as well- a GUI, more complex generation, multiple password generation, and more are all things that could be considered for the future.

User Stories

As a content creator, I want my passwords to be displayed in order so I know which passwords are the oldest and which are the newest in order to keep organized.

As a business owner, I want my passwords to be associated with a keyword or name so that I know which password matches up with which account to avoid any confusion on which supplier website uses which password.

As an older user, I want to know if I'm generating a duplicate password for a cite to avoid making any unnecessary changes and negate confusion.

As a user with specific password requirements, I want to customize the password length and characters used so that it meets the criteria of different websites.

As a non-technical user, I want clear explanations of each step so I understand what I'm choosing and why it matters.

As a team leader, I want to generate multiple unique passwords at once so I can assign them to team members or accounts.

Technical Requirements

Python and IDE: This program will be constructed within the most up to date version of PyCharm. It will use standard coding notation through notes to allow an easy understanding of what a given line of code does.

Variables and Data Types: This program will take input in the form of strings, use random integers for password generation.

Operations: TBD – This program may ask the user for simple number input, multiple, divide, or add, then use the produced number in the password somewhere.

Boolean Operations and Conditionals: This program will use utilize if/else/elif statements in order to generate passwords. Specifically, else statements will be used to handle errors in user input, elif statements to proceed, and more.

Loops: A simple loop will be implemented at the end of the program to prompt the user on if they wish to generate another password, if not, “no” will stop the program.

Data structures: Lists will be used store newly generated passwords, dictionaries may be used in order to connect generated passwords to the websites they’re associated with.

Code Modularization: The code will be broken down into no less than 3 functions, one for password generation, one to connect generated passwords to input from the user(websites in this case), and another to store the generated passwords and connections in a text document.

Task List

The top milestones for this project will be setting up the major functions, finding the best ways to prompt the user for good, memorable input, and making sure to reduce or completely eradicate any redundancies in code. Getting each of the main 3 functions to work may take anywhere from 1 to 3 days each. On the high end it may take a week each. After the functions are complete the following parts, those being good input and reduction in redundancy should be done in no more than a single week.

Citation

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